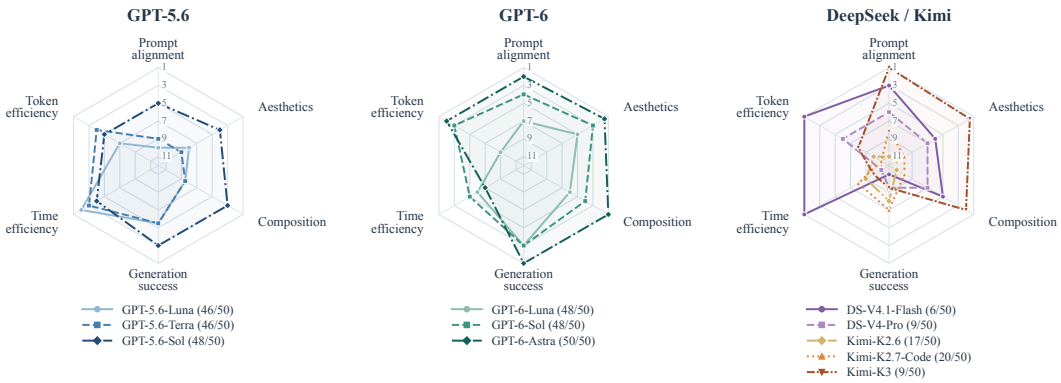


Figure 2: **Model rankings on MaLiang-IBench.** Ranks are computed across 11 models; better ranks lie farther from the center, with ties averaged. GPT models achieve 92–100% generation success, versus 12–40% for DeepSeek and Kimi. Mean quality scores use successful outputs (counts in parentheses). Cost ranks favor less time and fewer tokens.

GPT-5.6-Sol. These percentages use the full task set of each benchmark as the denominator. These results distinguish successful execution from satisfaction of visual requirements. Fig. 2 compares 11 models on MaLiang-IBench across visual quality, generation success, and computational cost, revealing different model rankings across these dimensions. Fig. 1 complements these comparisons with examples of image generation, visual understanding and editing, and video generation.

Our main contributions are:

- We introduce and define the **Program-to-Visual (P2V) gap** as the discrepancy between program-level correctness and visual requirement satisfaction, motivating a stateful formulation of visual program generation through construction, inspection, and revision.
- We introduce **MaLiang-Harness**, a unified framework for programmable image and video generation. Persistent Executable Generation state, a Traceable Generation Process, and Revision-aware Editing and Verification support continued refinement, inspection of construction histories, and verification of the current output.
- We evaluate 11 MLLMs on MaLiang-IBench and four on MaLiang-VBench, characterizing differences in generation success, visual quality, and computational cost. Comparison with public general-capability scores shows that similar benchmark performance can correspond to substantially different visual generation outcomes.

2 Related Work

Image and Video Generation. Diffusion models synthesize images through learned denoising processes [Ho et al., 2020], with latent diffusion reducing the cost of high-resolution synthesis [Rombach et al., 2022]. Flow matching provides a related framework for learning continuous generative dynamics [Lipman et al., 2022]. Video Diffusion Models and Stable Video Diffusion extend learned visual synthesis to temporal content [Ho et al., 2022, Blattmann et al., 2023]. These approaches also support explicit conditioning: ControlNet, for example, introduces spatial guidance through inputs such as edges and depth [Zhang et al., 2023]. Our focus is on the representation through which visual content is constructed and revised. MaLiang-Harness maintains an executable artwork whose geometry and temporal behavior can be inspected and edited.

Executable Visual Representations. Prior work establishes several ways to connect language models with executable visual representations. VISPROG generates modular programs for visual reasoning and image editing, exposing intermediate results as inspectable rationales [Gupta and Kembhavi, 2023]. Design2Code evaluates the translation of webpage screenshots into renderable implementations [Si et al., 2025]. In three-dimensional graphics, BlenderAlchemy combines a vision-based edit generator with a state evaluator to search for edits that realize a user's design intent [Huang et al., 2024]. These studies demonstrate that programs can mediate visual understanding and